In the Name of God



Probability and Statistics in Engineering Project

Multi-Armed Bandit

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Theory Question 1

Example 1: Clinical trial (two new drugs and one standard drug)

- **a) Arms:** The three treatments: new drug A, new drug B and the standard drug. Each has an unknown success probability (or mean improvement).
- **b) Reward:** The outcome for the treated patient, e.g. 1 if the patient recovers (or responds to treatment) and 0 otherwise. A continuous measure, such as the reduction of a symptom score, can also be used.
- **c) Round:** One patient (or one group of patients) enters the trial and receives one of the treatments; t counts the patients.
- **d) Exploitation and exploration:** Exploitation means giving the next patient the treatment that has worked best so far. Exploration means also giving the other treatments to some patients, so that their effectiveness can be measured reliably.
- **Why both are needed:** With only exploitation, an unlucky start (e.g. the best drug fails on its first few patients) can lead the trial to drop the best drug and treat everyone with an inferior one. With only exploration (a fixed equal allocation), many patients keep receiving a treatment that is already known to be worse, which is both costly and ethically problematic.

Example 2: Advertisements

- **a) Arms:** The different banners that can be shown.
- **b) Reward:** Whether the user clicks the banner (1 for a click, 0 otherwise), or the revenue from that view. The mean reward of a banner is its click-through rate.
- **c) Round:** One page view, at which a single banner is displayed to a visitor.
- **d) Exploitation and exploration:** Exploitation means showing the banner with the highest observed click-through rate. Exploration means showing other banners, especially new or rarely shown ones, to estimate their click-through rates.
- **Why both are needed:** With only exploitation, the site can lock onto a banner that happened to get a few early clicks and never discover a better one. With only exploration, the views are spread evenly over all banners, including clearly poor ones, and many clicks (and revenue) are lost.

Example 3: Movie recommendations

- **a) Arms:** The candidate movies, or movie categories, that can be recommended to the user.
- **b) Reward:** The user's feedback: whether the recommended movie is watched to the end, the watch time, or an explicit rating or like.
- **c) Round:** One evening (one recommendation) on which the platform suggests a movie to the user.

- **d) Exploitation and exploration:** Exploitation means recommending the kind of movie the user has liked most so far. Exploration means recommending movies or categories the user has rarely seen, to learn whether they would like them.
- **Why both are needed:** With only exploitation, the system keeps recommending one category and never learns that the user might like another one better (a "filter bubble"). With only exploration, the user mostly receives random suggestions that do not match their taste and may stop using the service.

Theory Question 2

Based on the problem description, we model the scenario as a Multi-Armed Bandit problem:

- **a) Arm:** Movie Genre (we want to maximize the rating over T movie-watching rounds/days).
- **b) Reward:** Rating (taking discrete values from 0 to 5).
- **c) Round:** The number of times the student watches a movie during the semester.
- **d) 1. Exploitation:** Choosing the genre that has received the highest average rating so far.
- **d) 2. Exploration:** Diversifying selections by choosing less-watched genres.

A purely exploitative strategy would be: if the Action genre receives the highest rating in the first 5 rounds, we exclusively watch Action movies thereafter. However, the student might eventually get tired of this genre, and we might ignore other untested genres that could potentially yield even higher ratings, thus failing to maximize overall satisfaction. A purely exploratory strategy involves constantly trying new or unseen genres. Obviously, not all genres will be appealing, leading to suboptimal total ratings over the semester.

Intuitively, an interactive, alternating approach between exploration and exploitation is required. For example, exploring for T' rounds, then exploiting the best genre for T'' rounds, and subsequently returning to exploration if the ratings for the exploited genre begin to decline over time.

Theory Question 3

If we know that the i-th arm is chosen at round t, and it is the s-th time this arm is pulled, we have:

$$\mathbb{E}[r_t | A_t = i] = \mathbb{E}[X_{i,s}] = \mu_i$$

Because if the i-th arm is selected, our reward is $X_{i,s}$. Suppose T_r is the set of indices t where the indicator function is 1, meaning $A_t = i$. Then:

$$N_i(T) = \sum_{t=1}^{T} \mathbb{I}\{A_t = i\} = \sum_{t \in T_r} 1 + \sum_{t \notin T_r} 0 = |T_r|$$

This definition is intuitively logical because the size of the set T_r represents the total number of times the indicator A_t equals 1 (i.e., arm i is chosen).

From the previous equations, the expected cumulative reward is:

$$\mathbb{E}[S_T] = \mathbb{E}\left[\sum_{t=1}^{T} r_t\right] = \sum_{t=1}^{T} \mathbb{E}[r_t]$$

For the expected value of r_t:

$$\mathbb{E}[r_t] = \sum_{i=1}^{k} \mathbb{E}[r_t | A_t = i] \times P(A_t = i)$$

$$\mathbb{E}[N_i(T)] = \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{I}\{A_t = i\}\right] = \sum_{t=1}^{T} \mathbb{E}[\mathbb{I}\{A_t = i\}] = \sum_{t=1}^{T} P(A_t = i)$$

Therefore, we obtain:

$$\mathbb{E}[S_T] = \sum_{t=1}^{T} \sum_{i=1}^{k} \mu_i \times P(A_t = i) = \sum_{i=1}^{k} \mu_i \mathbb{E}[N_i(T)]$$

Theory Question 4

$$\mathbb{E}[\hat{\mu}_{i,n}] = \mathbb{E}\left[\frac{1}{n}\sum_{s=1}^{n} X_{i,s}\right] = \frac{1}{n}\sum_{s=1}^{n} \mathbb{E}[X_{i,s}] = \mu_i$$

$$\mathrm{Var}(\hat{\mu}_{i,n}) = \mathrm{Var}\left(\frac{1}{n}\sum_{s=1}^{n} X_{i,s}\right) = \frac{1}{n^2}\sum_{s=1}^{n} \mathrm{Var}(X_{i,s}) = \frac{\mu_i(1-\mu_i)}{n}$$

By Chebyshev's inequality, for any $\varepsilon > 0$:

$$P\left(|\hat{\mu}_{i,n} - \mu_i| \geq \varepsilon\right) \leq \frac{\mathrm{Var}(\hat{\mu}_{i,n})}{\varepsilon^2} = \frac{\mu_i(1-\mu_i)}{n\varepsilon^2}$$

This relation holds for any $\varepsilon \in \mathbb{R}^+$. As n grows arbitrarily large:

$$0 \leq \lim_{n\to\infty} P\left(|\hat{\mu}_{i,n} - \mu_i| \geq \varepsilon\right) \leq \lim_{n\to\infty} \frac{\mu_i(1-\mu_i)}{n\varepsilon^2} = 0$$

$$\implies \lim_{n\to\infty} P\left(|\hat{\mu}_{i,n} - \mu_i| \geq \varepsilon\right) = 0$$

Thus, the sample mean converges in probability to the true mean (the weak law of large numbers).

Theory Question 5

$$\mathbb{E}[S_T] = \sum_{i=1}^{k} \mu_i \mathbb{E}[N_i(T)]$$

The number of times the optimal arm is pulled in an ideal scenario is:

$$N_{i^*}(T) = \sum_{t=1}^{T} \mathbb{I}\{A_t = i^*\} = \sum_{t=1}^{T} 1 = T$$

Hence, the Regret for always picking the optimal arm is:

$$\text{Regret}(T) = T\mu^* - \mu_{i^*} T = 0 \quad \text{for} \quad T \in \mathbb{N}$$

Also:

$$\sum_{i=1}^{k} \mathbb{E}[N_i(T)] = \mathbb{E}\left[\sum_{i=1}^{k} N_i(T)\right] = \mathbb{E}\left[\sum_{i=1}^{k}\sum_{t=1}^{T} \mathbb{I}\{A_t = i\}\right] = \mathbb{E}\left[\sum_{t=1}^{T} 1\right] = T$$

$$\implies \text{Regret}(T) = \sum_{i=1}^{k} \mu^* \mathbb{E}[N_i(T)] - \sum_{i=1}^{k} \mu_i \mathbb{E}[N_i(T)] = \sum_{i=1}^{k} \Delta_i \mathbb{E}[N_i(T)]$$

Based on the equation above, Δ_i is independent of the bandit algorithm used for any $1 \leq i \leq k$. Thus, to analyze regret, we must look at the sum of expected selections of suboptimal arms.

Simulation Question 1

Averaged over the 500 runs, the regret at $T = 1000$ is 300.0 for the uniform policy and 56.3 for the greedy policy. The uniform curve is a straight line that lies on the theoretical value $\frac{t}{k}\sum_i \Delta_i = \frac{t}{3}(0.6+0.3+0) = 0.3\,t$ (dashed line), as Theory Question 6 predicts. The greedy strategy performs much better on average because it uses the observed rewards, but its regret also grows linearly. After one pull per arm it can lock onto a sub-optimal arm, because an arm whose empirical mean drops to 0 is never tried again. This happened in 16% of the runs. Those runs lose 0.3 or 0.6 per step for the rest of the horizon, which gives the average curve a constant positive slope. Neither policy reaches the sub-linear regret that a good bandit algorithm should have.

Theory Question 6

Given the uniform policy, the indicator variable $\mathbb{I}\{A_t = i\}$ follows:

$$\mathbb{I}\{A_t = i\} \sim \text{Ber}\left(\frac{1}{k}\right)$$

Therefore:

$$N_i(T) = \sum_{t=1}^{T} \mathbb{I}\{A_t = i\} \sim \text{Bin}\left(T, \frac{1}{k}\right)$$



Figure 1: Regret Comparison: Uniform vs Greedy Strategy ($N = 500$ runs)

$$\text{Regret}(T) = \sum_{i=1}^{k} \Delta_i \mathbb{E}[N_i(T)] = \sum_{i=1}^{k} \Delta_i \mathbb{E}\left[\text{Bin}\left(T, \frac{1}{k}\right)\right] = \frac{T}{k} \sum_{i=1}^{k} \Delta_i$$

Using the given parameters, the regret is:

$$\text{Regret}(1000) = \frac{1000}{2}(0 + 0.3) = 150$$

Obviously, if we strictly followed a policy that only chose the optimal arm, the regret would be zero. The uniform strategy is sub-optimal because observing the result of an arm pull provides information about that arm's parameter. As information accumulates, a rational agent should lean towards finding and exploiting the optimal arm. The uniform policy acts completely blind and ignores this incoming data.

Theory Question 7

Let $T = 1000$ for the numerical illustrations, with $\mu_1 = 0.9$ and $\mu_2 = 0.6$, so $\Delta_2 = 0.3$.

1. Policy A (pure exploitation). Policy A decides after a single observation of each arm, and a single Bernoulli observation says very little about the mean. With probability $0.1 \times 0.6 = 0.06$ the better arm gives 0 and the worse arm gives 1. With probability $0.9 \times 0.6 + 0.1 \times 0.4 = 0.58$ the two rewards are equal and the choice is a coin flip. So the policy commits to the worse arm with probability $0.06 + 0.29 = 0.35$, and it never corrects the mistake. In that case it loses $\Delta_2 = 0.3$ at every remaining step, a regret of about $0.3\,T$. Its expected regret is $0.3 + 0.35 \times 0.3 \times (T - 2) \approx 0.105\,T \approx 105$, which is still linear in T.

2. Policy B (uniform exploration) compared with A. By Theory Question 6, Policy B has regret exactly $\frac{T}{2}\Delta_2 = 0.15\,T = 150$. It is not always better than A. In about

65% of the runs, A picks the right arm and its regret is only 0 or 0.3, far below 150. In this example, A's *expected* regret (≈ 105) is even lower than B's. A's danger is its variance: in about 35% of the runs its regret is about 300, twice B's. Both policies have linear regret: B because it never stops exploring, A because it stops exploring too early.

3. Policy C (explore 10%, then exploit). During the first $0.1\,T$ steps, Policy C pays the uniform price of $0.1\,T \times 0.15 = 15$. By the end of that phase each arm has been pulled about 50 times. The standard deviation of the difference of the two empirical means is then about $\sqrt{(0.09 + 0.24)/50} \approx 0.08$. A gap of 0.3 is therefore about 3.7 standard deviations, and the best arm is identified with probability above 0.999. The greedy phase also keeps updating the empirical means, so the rare early mistake can still be corrected. The expected regret is about 15, much smaller than for A or B. C balances the two goals by exploring while the estimates are unreliable and exploiting once they are accurate. Its exploration phase grows in proportion to T, so its regret is still linear, only with a small slope. UCB and Thompson Sampling go further: they reduce exploration adaptively and reach $O(\log T)$ regret.

Theory Question 8

According to Hoeffding's inequality for $c_n(\delta)$:

$$P(|Y_n - \mu| \geq c_n(\delta)) \leq 2\exp\left(-2n\left(\sqrt{\frac{\ln(2/\delta)}{2n}}\right)^2\right) = \delta$$

$$\implies P(|Y_n - \mu| < c_n(\delta)) = 1 - P(|Y_n - \mu| \geq c_n(\delta)) \geq 1 - \delta$$

(Here Y_n denotes the sample mean $\bar{Y}_n$, and μ denotes m.) **Interpretation as a confidence interval:** the probability that the *random* interval $[\bar{Y}_n - c_n(\delta), \bar{Y}_n + c_n(\delta)]$ contains the fixed true mean m is *at least* $1 - \delta$. Its half-width $c_n(\delta)$ shrinks as $1/\sqrt{n}$ and grows only logarithmically as δ decreases. With $\delta = 2t^{-4}$, $\ln(2/\delta) = 4\ln t$, so $c_n = \sqrt{4\ln t/(2n)} = \sqrt{2\ln t/n}$, which is the radius $c_i(t)$ used by UCB. In a pure exploitation strategy, we seek to maximize $\mu_{i,t}$. However, exploration is crucial in early stages. The Upper Confidence Bound expression $\mu_{i,t} + c_{i,t}$ perfectly satisfies both needs. Because: a) An arm that has been played only a few times has a small $N_i(t)$ and therefore a large bonus $c_i(t)$. Its upper bound $\hat{\mu}_i + c_i(t)$ can exceed that of the current leader even if its empirical mean is lower, so it gets tried again (exploration). Because $c_i(t)$ grows with $\ln t$, an arm that is neglected for a long time eventually becomes attractive again. b) Each pull of arm i increases $N_i(t)$ and shrinks $c_i(t)$ like $1/\sqrt{N_i(t)}$. Well-sampled arms are then judged mostly by their empirical mean (exploitation). A sub-optimal arm stops being chosen once its whole confidence interval falls below μ^*, which requires only about $\frac{8\ln t}{\Delta_i^2}$ pulls. Hence, selecting the arm that maximizes $\mu_{i,t} + c_{i,t}$ creates an elegant and effective balance between exploration and exploitation.

Simulation Question 2

With $N = 10000$ repetitions per value of n:

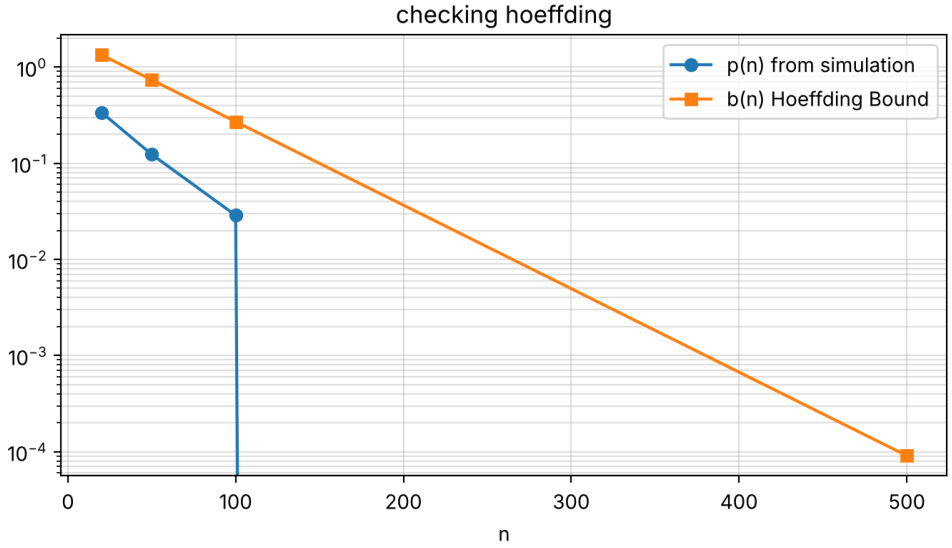


Figure 2: Checking Hoeffding Bound

n	$\hat{p}_n$ (simulation)	$b_n = 2e^{-2n(0.1)^2}$
20	0.3383	1.3406
50	0.1235	0.7358
100	0.0287	0.2707
500	0.0000	0.0001

As Hoeffding's inequality guarantees, $\hat{p}_n \leq b_n$ for every n. For $n = 20$ the bound exceeds 1 and is therefore trivial. Both quantities decrease exponentially as n grows, so the sample mean concentrates around 0.7. The bound is conservative, however: it is 4–10 times larger than the true probability. This is because it holds for *every* distribution on $[0, 1]$, including the worst case (variance $1/4$), while this Bernoulli variable has variance $0.7 \times 0.3 = 0.21$. At $n = 500$, the event $|\bar{Y}_n - 0.7| \geq 0.1$ lies about 4.9 standard deviations away, so its probability ($\approx 10^{-6}$) is too small to observe in 10^4 trials. The plot uses a logarithmic axis, so this zero is not drawn.

Theory Question 9

$$P(\mu_i) = \text{Beta}(\alpha_i, \beta_i) \propto \mu_i^{\alpha_i - 1}(1 - \mu_i)^{\beta_i - 1}$$

$$P(\mu_i|H_t) \propto P(H_t|\mu_i)P(\mu_i) \propto \mu_i^{\alpha_i - 1}(1 - \mu_i)^{\beta_i - 1}\mu_i^S(1 - \mu_i)^f \propto \mu_i^{\alpha_i + S - 1}(1 - \mu_i)^{\beta_i + f - 1}$$

$$\implies P(\mu_i|H_t) = \text{Beta}(\alpha_i + S, \beta_i + f)$$

As shown, the structural form of the posterior distribution is identical to the prior. This proves that the Beta distribution is the conjugate prior for the Bernoulli likelihood. The primary advantage of a conjugate prior is algebraic convenience, completely bypassing the need for complex integral calculations.

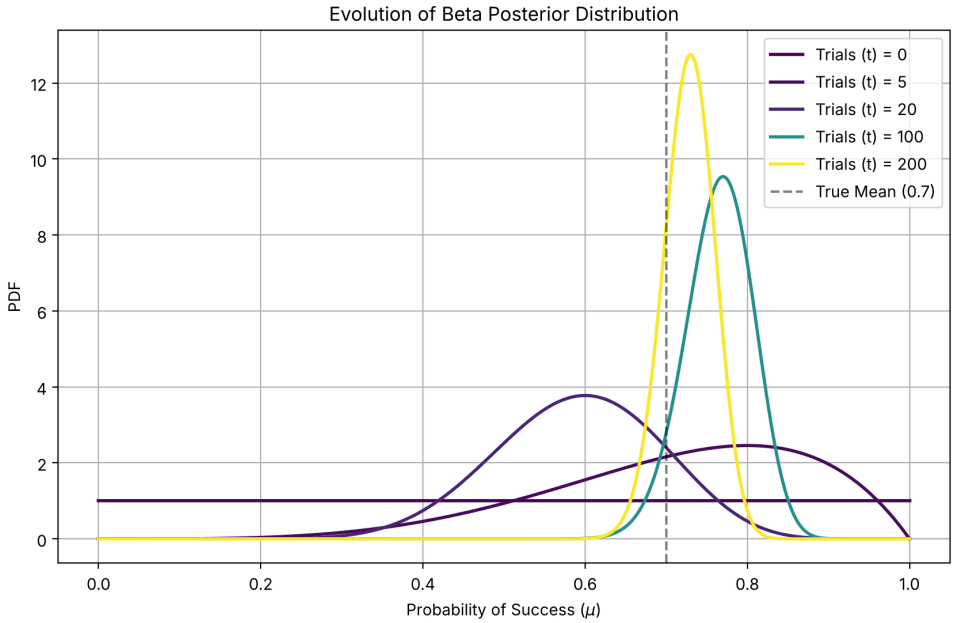


Figure 3: Evolution of Beta Posterior Distribution

Simulation Question 3

At $t = 0$ the posterior is the uniform prior Beta$(1, 1)$. After t observations with s successes it is Beta$(1 + s,\ 1 + t - s)$. Its mean $\frac{1+s}{2+t}$ moves to the observed success rate, which tends to 0.7 by the law of large numbers. Its standard deviation falls roughly as $\sqrt{0.7 \times 0.3/t}$, about 0.03 at $t = 200$. So the PDF becomes narrower and taller and concentrates around the true mean 0.7 (dashed line). The early curves ($t = 5$) can still sit visibly away from 0.7, because a few observations carry a lot of randomness.

Theory Question 10

Let the data consist of $n = s + f$ independent Bernoulli(μ) trials with s successes and f failures. The likelihood of a given sequence is $\mu^s(1-\mu)^f$. If only the counts are recorded it is $\binom{s+f}{s}\mu^s(1-\mu)^f$; the binomial coefficient does not depend on μ. By Bayes' rule, with the Beta(α, β) prior:

$$P(\mu \mid \text{data}) \propto P(\text{data} \mid \mu)\, P(\mu) = \binom{s+f}{s}\mu^s(1-\mu)^f \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\,\mu^{\alpha-1}(1-\mu)^{\beta-1}.$$

All factors that do not depend on μ are absorbed into the proportionality constant:

$$P(\mu \mid \text{data}) \propto \mu^{\alpha+s-1}(1-\mu)^{\beta+f-1}, \qquad 0 \le \mu \le 1.$$

This is the kernel of Beta$(\alpha + s,\ \beta + f)$. Since the posterior must integrate to 1, the normalizing constant is fixed, and

$$P(\mu \mid \text{data}) = \frac{\Gamma(\alpha+\beta+s+f)}{\Gamma(\alpha+s)\,\Gamma(\beta+f)}\,\mu^{\alpha+s-1}(1-\mu)^{\beta+f-1},$$

that is, $\alpha' = \alpha + s$ and $\beta' = \beta + f$.

Simulation Question 4

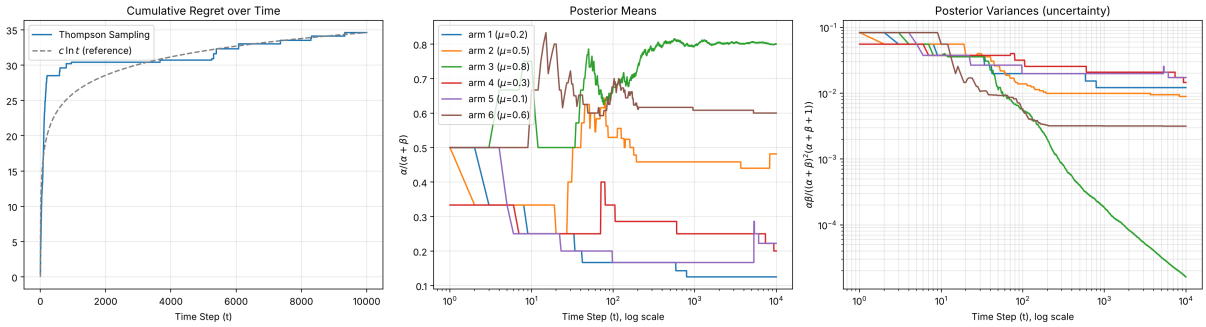


Figure 4: Cumulative Regret and Convergence of Posterior Means

The bandit has six Bernoulli arms with means $(0.2, 0.5, 0.8, 0.3, 0.1, 0.6)$, run for $T = 10000$ steps with Beta$(1, 1)$ priors.

Regret (left). Almost all of the regret (≈ 30 of the final 34.6) is paid in the first few hundred steps. After that the curve grows only in rare, small jumps, which matches the $O(\log T)$ behaviour (dashed $c \ln t$ reference).

Posterior means (middle). The best arm (0.8) received 9881 of the 10000 pulls, and its posterior mean converges to 0.800. The second-best arm (0.6) was pulled 73 times and is also well estimated. The weak arms were pulled only 6–25 times each, so their estimates are still rough: 0.125 for $\mu = 0.2$, and 0.222 for $\mu = 0.1$. This is expected: once an arm is clearly worse, Thompson Sampling stops spending pulls on it, and an accurate estimate of a bad arm is not needed to avoid it.

Posterior variances (right). The variance of the best arm falls like $1/N$, down to about 2×10^{-5} (standard deviation 0.004). The variances of the other arms stop decreasing as soon as those arms stop being played. Their uncertainty stays large but no longer matters, because their posteriors lie well below that of the best arm.

Theory Question 11

In the Beta-Bernoulli Bayesian model, after observing the data, our belief distribution regarding the reward parameter is updated to a Beta posterior distribution Beta(α', β'). The Probability Density Function (PDF) of this distribution is defined as follows:

$$f(\mu; \alpha', \beta') = \frac{1}{B(\alpha', \beta')} \mu^{\alpha'-1}(1-\mu)^{\beta'-1}$$

where $B(\alpha', \beta')$ is the Beta function, calculated as:

$$B(\alpha', \beta') = \int_0^1 t^{\alpha'-1}(1-t)^{\beta'-1} dt = \frac{\Gamma(\alpha')\Gamma(\beta')}{\Gamma(\alpha' + \beta')}$$

To calculate the expected value (mean) of the random variable, we have:

$$\mathbb{E}[\mu] = \int_0^1 \mu \cdot f(\mu; \alpha', \beta') d\mu$$

Substituting the density function into the integral:

$$\mathbb{E}[\mu] = \int_0^1 \mu \cdot \frac{\mu^{\alpha'-1}(1-\mu)^{\beta'-1}}{B(\alpha',\beta')} d\mu = \frac{1}{B(\alpha',\beta')} \int_0^1 \mu^{\alpha'}(1-\mu)^{\beta'-1} d\mu$$

The term inside the integral is the definition of the Beta function with parameters $\alpha'+1$ and β':

$$\mathbb{E}[\mu] = \frac{B(\alpha'+1,\beta')}{B(\alpha',\beta')}$$

Using the relationship between the Beta and Gamma functions, we expand the above expression:

$$\mathbb{E}[\mu] = \frac{\frac{\Gamma(\alpha'+1)\Gamma(\beta')}{\Gamma(\alpha'+1+\beta')}}{\frac{\Gamma(\alpha')\Gamma(\beta')}{\Gamma(\alpha'+\beta')}} = \frac{\Gamma(\alpha'+1)}{\Gamma(\alpha')} \cdot \frac{\Gamma(\alpha'+\beta')}{\Gamma(\alpha'+\beta'+1)}$$

Using the recursive property of the Gamma function, i.e., $\Gamma(z+1) = z\Gamma(z)$, we will have:

$$\Gamma(\alpha'+1) = \alpha'\Gamma(\alpha') \quad \text{and} \quad \Gamma(\alpha'+\beta'+1) = (\alpha'+\beta')\Gamma(\alpha'+\beta')$$

Substituting these values into the mean equation:

$$\mathbb{E}[\mu] = \frac{\alpha'\Gamma(\alpha')}{\Gamma(\alpha')} \cdot \frac{\Gamma(\alpha'+\beta')}{(\alpha'+\beta')\Gamma(\alpha'+\beta')}$$

Finally, by simplifying the similar terms, the mean of the posterior distribution is obtained:

$$\mathbb{E}[\mu] = \frac{\alpha'}{\alpha'+\beta'}$$

Theory Question 12

To calculate the probability of success in the next trial given past data, we use the Posterior Predictive Distribution. This probability is obtained by averaging over the parameter:

$$P(X_{t+1} = 1|\text{data}) = \int_0^1 P(X_{t+1} = 1|\mu) f(\mu|\text{data}) d\mu$$

Given that $P(X_{t+1} = 1|\mu) = \mu$ and our posterior distribution is $f(\mu|\text{data}) = \text{Beta}(\alpha,\beta)$, we have:

$$P(X_{t+1} = 1|\text{data}) = \int_0^1 \mu \frac{\mu^{\alpha-1}(1-\mu)^{\beta-1}}{B(\alpha,\beta)} d\mu$$

This integral represents the expected value of the $\text{Beta}(\alpha,\beta)$ distribution. Therefore:

$$P(X_{t+1} = 1|\text{data}) = \frac{\alpha}{\alpha+\beta}$$

Analysis in the Context of Sequential Decision Making: In sequential decision-making processes like Multi-Armed Bandits, this result shows that our best prediction for the future success rate of an arm is exactly the mean of our current belief distribution. This value acts as a criterion for selecting arms in greedy algorithms. As the number of observations increases (increasing the weight $\alpha+\beta$), the real data heavily outweighs the prior beliefs, and our prediction converges closer to the arm's true physical reality.

Theory Question 13

Data: First arm $\sim$ Beta$(3, 2)$, second arm $\sim$ Beta$(10, 5)$.

Calculations: Means:

$$\mathbb{E}[\mu_1] = \frac{3}{3+2} = 0.6 \quad , \quad \mathbb{E}[\mu_2] = \frac{10}{10+5} \approx 0.667$$

Variances:

$$\mathrm{Var}(\mu_1) = \frac{3 \times 2}{(3+2)^2(3+2+1)} = \frac{6}{150} = 0.04 \quad \text{(sd 0.20)}$$

$$\mathrm{Var}(\mu_2) = \frac{10 \times 5}{(10+5)^2(10+5+1)} = \frac{50}{3600} \approx 0.0139 \quad \text{(sd 0.12)}$$

Thompson Sampling chooses arm 1 exactly when $\theta_1 > \theta_2$. Numerical integration gives

$$P(\theta_1 > \theta_2) = \int_0^1 f_{\mathrm{Beta}(3,2)}(x)\, F_{\mathrm{Beta}(10,5)}(x)\, dx \approx 0.41.$$

So TS picks arm 2 about 59% of the time and arm 1 about 41% of the time.

Thompson Sampling Behavior:

1. **Exploitation:** The posterior mean of the second arm is higher ($0.667 > 0.6$), and its posterior is narrow (it is based on about 13 observations instead of 3), so TS selects the second arm more often.

2. **Exploration:** The first arm has almost three times the variance (0.04 against 0.0139). Its wide posterior often produces large samples: $P(\theta_1 > 0.8) \approx 0.18$, against 0.13 for arm 2. So, despite its lower mean, it is still chosen in about 41% of the steps.

Balance: The difference in means is small compared with the uncertainty about arm 1, so TS keeps testing arm 1 often. Each test narrows its posterior. If arm 1 really is worse, its probability of being chosen then falls quickly. A greedy rule would pick arm 2 every time and never learn whether arm 1 is actually better.

Simulation Question 5

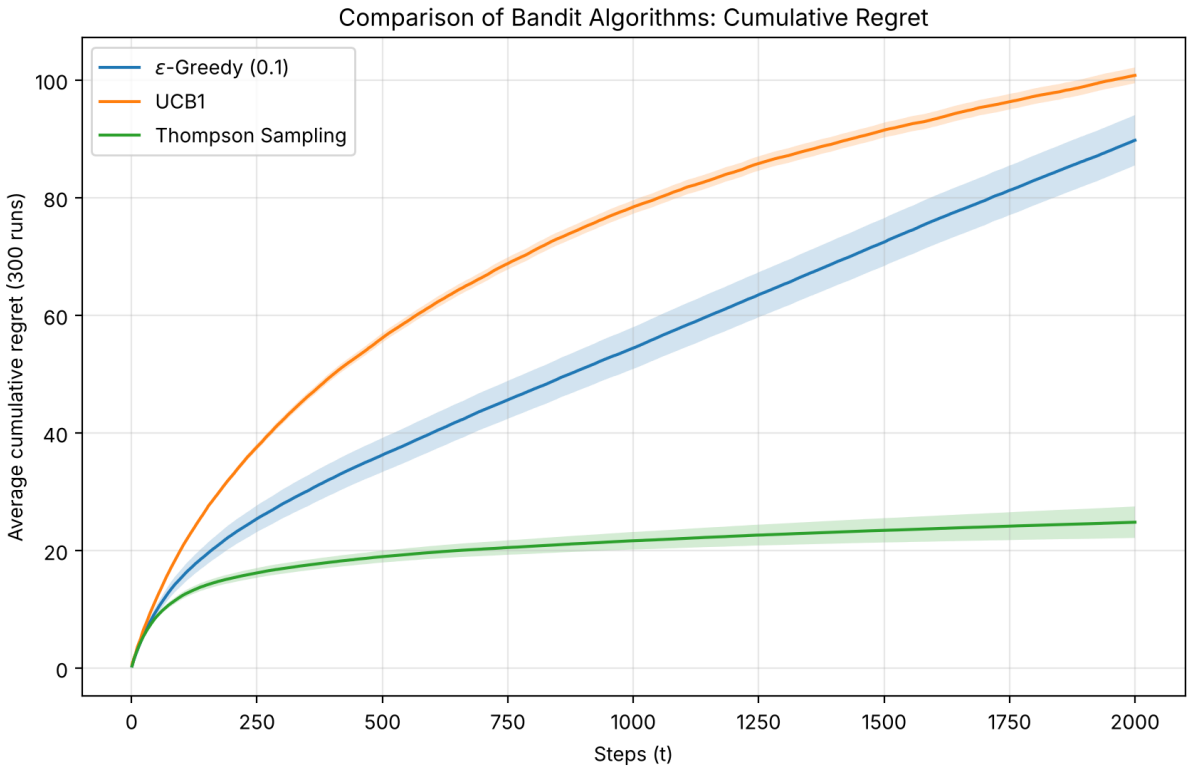


Figure 5: Comparison of Cumulative Regret: ϵ-Greedy, UCB, and Thompson Sampling

The curves are averages over 300 independent runs (shaded: 95% confidence band). In every run the three algorithms face the same table of random rewards, so the comparison is not affected by luck in the rewards. Averaged regret at $T = 2000$:

Algorithm	Regret at $T = 2000$
ϵ-Greedy ($\epsilon = 0.1$)	89.7 ± 4.3
UCB1	100.8 ± 1.3
Thompson Sampling	24.8 ± 2.7

Thompson Sampling has by far the lowest regret. ϵ-Greedy keeps exploring at a fixed rate, so after a short learning phase its regret grows linearly, with slope $\epsilon \cdot \frac{1}{k}\sum_i \Delta_i \approx 0.034$ per step. UCB1 grows only logarithmically, but its constant is large: the bonus $\sqrt{2 \ln t / N_{i,t}}$ keeps the arms with means 0.5 and 0.3 in play for a long time. That is why UCB1 is still above ϵ-Greedy at $T = 2000$. A longer run (40 runs, $T = 20000$) shows the crossing: at $T = 5000$ the regrets are 132 (UCB1) and 191 (ϵ-Greedy), and at $T = 20000$ they are 178 and 703, against 34 for Thompson Sampling.

Theory Question 14

The Thompson Sampling (TS) algorithm achieves a dynamic balance between exploration and exploitation due to its Bayesian approach. The reasons for achieving $\mathcal{O}(\log T)$ regret

can be summarized as follows:

1. **Adaptivity:** In TS, the probability of choosing a sub-optimal arm decreases exponentially as the number of observations increases. Unlike algorithms with a fixed exploration rate, TS only explores when there is uncertainty regarding the optimality of an arm.

2. **Posterior Concentration:** As time T increases, the variance of the posterior distribution for each arm approaches zero (for arm i the posterior variance $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ falls roughly as $\mu_i(1-\mu_i)/N_i(t)$). This concentration severely limits the probability of drawing large random samples from sub-optimal distributions, which would otherwise lead to choosing the wrong arm.

3. **Matching the Lai-Robbins Bound:** It has been proven that the expected number of times a sub-optimal arm i is chosen in the TS algorithm asymptotically satisfies:
$$\mathbb{E}[N_i(T)] \leq (1+\epsilon)\frac{\log T}{KL(\mu_i||\mu^*)}$$
where KL is the Kullback-Leibler divergence between the distribution of the sub-optimal arm and the optimal arm. Since the total regret is derived from the relation $R(T) = \sum \Delta_i \mathbb{E}[N_i(T)]$, the final regret will be of the order $\mathcal{O}(\log T)$.

Therefore, Thompson Sampling is considered an Asymptotically Optimal algorithm that achieves the most optimal possible learning rate.

Theory Question 15

The Probability Matching principle states that an agent should select an arm with a probability equal to its belief that the arm is optimal. If θ_j is the unknown parameter of arm j and H is the history of observations, this probability is:

$$\pi_i = P(\theta_i = \max_j \theta_j | H) = \int_\Theta \mathbb{I}(\theta_i = \max_j \theta_j) P(\theta|H) d\theta$$

Assuming the posterior independence of arms, we have $P(\theta|H) = \prod_{j=1}^{K} P(\theta_j|H)$. Explicitly calculating this multiple integral is highly complex.

The Role of Thompson Sampling as a Monte Carlo Method: Thompson Sampling solves this integral using the Monte Carlo method with just a single sample ($n = 1$) at each time step:

1. For each arm, a sample $\hat{\theta}_j$ is drawn from the posterior distribution $P(\theta_j|H)$.

2. The arm $A_t = \arg\max_j \hat{\theta}_j$ is selected.

With this single draw, the probability that arm i is chosen is exactly π_i, so Thompson Sampling implements probability matching exactly, not approximately. In other words, the probability that the random sample of one arm is larger than the rest in a given time step is exactly equal to the value of the integral. Thus, Thompson Sampling automatically implements probability matching without requiring heavy integration computations.

Simulation Question 6

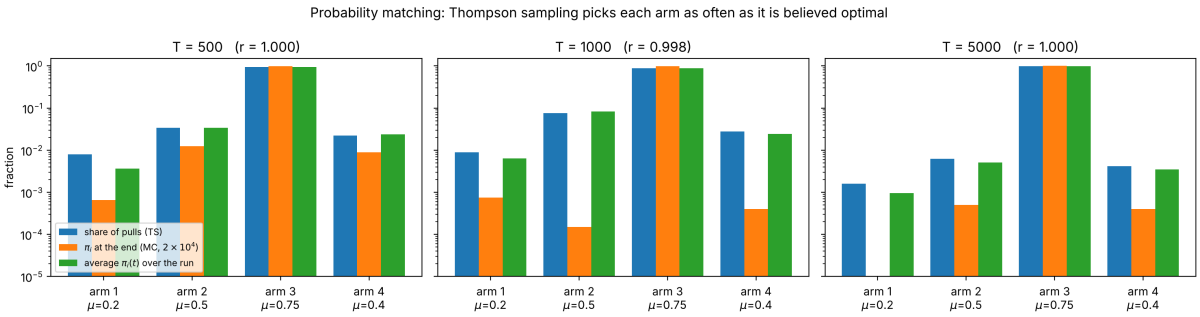


Figure 6: Share of pulls of each arm, posterior probability of being optimal at the end of the run, and its average over the run (log scale). The bandit has means $(0.2, 0.5, 0.75, 0.4)$.

Method. The function `prob_optimal` estimates $\pi_i = P(\theta_i = \max_j \theta_j \mid H)$ by Monte Carlo, as in Theory Question 15. It draws $n = 20000$ joint samples from the independent Beta posteriors and counts how often each arm has the largest sample. For each $T \in \{500, 1000, 5000\}$, Thompson Sampling is run for T steps. The run records how often each arm was pulled, π_i at the end of the run, and the Pearson correlation between the two vectors.

Results (one run per T):

T	share of pulls of the best arm	final π_{best}	correlation
500	0.936	0.978	0.9999
1000	0.887	0.999	0.9978
5000	0.988	0.999	1.0000

A single run is noisy, so the experiment was repeated 100 times for each T. The mean correlation rose from 0.9979 to 0.9981 to 1.0000, and the worst run from 0.934 to 0.844 to 0.9998. The largest gap between the share of pulls and π_i fell from 0.073 to 0.044 to 0.011. So the agreement improves as T grows.

Why the correlation improves with T. The final π describes the posterior at the end of the run, while the share of pulls is an average over all T steps, including the early steps when the posterior was still uncertain. For small T, the uncertain phase is a large part of the run, so the two vectors differ: sub-optimal arms were pulled more often than their final π_i suggests. As T grows, the posteriors concentrate, $\pi_{\text{best}} \to 1$, and the early phase becomes a negligible share of the pulls. Both vectors then approach $(0, 0, 1, 0)$, and their correlation approaches 1. Note that both vectors are dominated by the best arm, so the correlation is close to 1 even for small T; the gaps on the log-scale plot show the differences more clearly.

A stricter check of probability matching. Probability matching states that, at every step, $P(A_t = i \mid H_t) = \pi_i(t)$. Summing over the run, the share of pulls should therefore equal the *average* of $\pi_i(t)$ over the same steps:

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{I}\{A_t = i\} \;\approx\; \frac{1}{T}\sum_{t=1}^{T} \pi_i(t).$$

To test this, the simulation also estimates $\pi(t)$ before every step, with 1000 Monte Carlo draws (green bars). The two sides agree to within 0.004, 0.006 and 0.002 for the three values of T, for every arm, including the rarely played ones. This confirms that Thompson Sampling implements probability matching at every step, not only in the limit.

Theory Question 16

The choice of the prior Beta distribution directly affects the balance between exploration and exploitation:

1. **Uniform Prior (**Beta$(1, 1)$**):** This distribution is equivalent to $P(\mu) = 1$. Due to its high variance ($\sigma^2 = 1/12$), it provides the maximum initial exploration. The algorithm has no initial bias and is updated purely based on observed rewards.

2. **Informative Prior (**Beta$(100, 100)$**):** This prior is equivalent to 200 imaginary observations per arm, half successes, and its standard deviation is only about 0.035. The posterior mean $\frac{100+s}{200+n}$ is pulled strongly towards 0.5, so it is biased until an arm has been played several hundred times. Because every arm starts with the same narrow posterior, the samples of different arms barely differ, and the first few hundred choices are close to uniform random. The effect is slow learning: the algorithm keeps spreading its pulls (a form of over-exploration) and is slow to commit to the best arm, which costs regret early on. If the prior is wrong (true means far from 0.5), this cost is large. If it is right, the prior saves exploration.

3. **Jeffreys Prior (**Beta$(0.5, 0.5)$**):** Theoretically, this distribution is invariant to reparameterization. The concentration of density at the boundaries 0 and 1 causes the very first observation to have a severe impact on the posterior distribution. This prior is suitable for quickly identifying extremely strong or extremely weak arms.

Conclusion: In the absence of prior knowledge, Beta$(1, 1)$ exhibits the most stable behavior, whereas Beta$(100, 100)$ keeps the estimates near 0.5 for a long time and delays the identification of the best arm.

Simulation Question 7

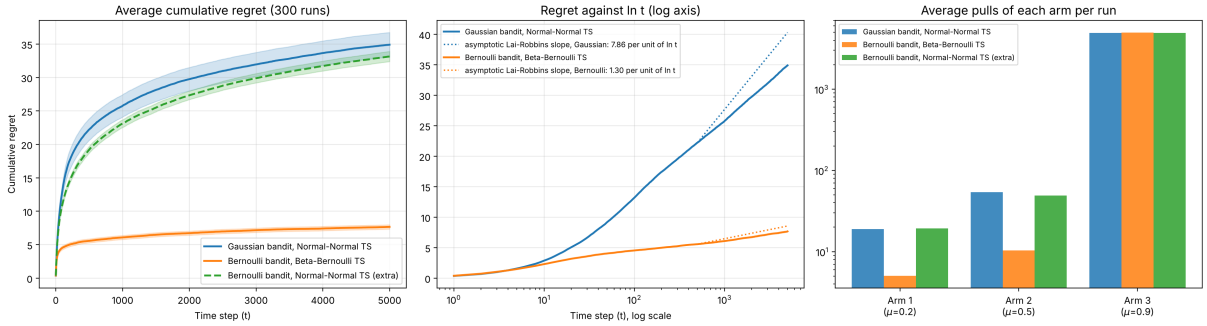


Figure 7: Thompson Sampling on a Gaussian bandit (Normal–Normal model) compared with the Bernoulli bandit (Beta–Bernoulli model), both with means $0.2, 0.5, 0.9$; averages over 300 runs.

Setup. The Gaussian bandit has three arms with rewards $X \sim \mathcal{N}(\mu_i, \sigma^2)$, where $\mu = (0.2, 0.5, 0.9)$ and $\sigma^2 = 1$ is known. Each mean gets the prior $\mathcal{N}(\hat{\mu}_0, \sigma_0^2) = \mathcal{N}(0, 1)$. At every step, $\theta_i \sim \mathcal{N}(\hat{\mu}_i, \sigma_i^2)$ is drawn for each arm and the arm with the largest θ_i is played. After observing reward X, the pulled arm is updated by the conjugate rule

$$\frac{1}{\sigma_1^2} = \frac{1}{\sigma_0^2} + \frac{1}{\sigma^2}, \qquad \hat{\mu}_1 = \sigma_1^2 \left(\frac{\hat{\mu}_0}{\sigma_0^2} + \frac{X}{\sigma^2} \right),$$

which is the formula of Section 6.9. After n pulls, the posterior variance is $1/(1/\sigma_0^2 + n/\sigma^2)$. For comparison, the Bernoulli case uses Bernoulli rewards with the same means and Beta–Bernoulli Thompson Sampling with a Beta$(1, 1)$ prior. Both run for $T = 5000$ steps, 300 times.

Results at $T = 5000$ (mean $\pm$ 95% confidence interval):

Case	Regret	Pulls (arms 1, 2)	Best arm
Gaussian bandit, Normal–Normal TS	34.9 ± 1.8	19.0, 54.0	98.5%
Bernoulli bandit, Beta–Bernoulli TS	7.7 ± 0.3	5.0, 10.4	99.7%
Extra: Bernoulli bandit, Normal–Normal TS	33.2 ± 0.8	19.4, 49.0	98.6%

Discussion. The Gaussian version works: its regret grows logarithmically and it plays the best arm 98.5% of the time. Its regret is, however, about 4.5 times that of the Bernoulli case. The reason is the noise level, not the algorithm. A Gaussian reward with $\sigma^2 = 1$ is much noisier than a Bernoulli reward, whose variance is $\mu(1-\mu) \leq 0.25$. In the Lai–Robbins bound this appears through the KL divergence:

- For Gaussian arms, $\mathrm{KL} = \Delta_i^2/(2\sigma^2)$, which is 0.08 for the arm with mean 0.5.
- For Bernoulli arms, $\mathrm{KL}(0.5 \,\|\, 0.9) = 0.51$.

The resulting constants $\sum_i \Delta_i / \mathrm{KL}_i$ are 7.86 (Gaussian) and 1.30 (Bernoulli). The Gaussian problem is therefore inherently about six times harder. The measured slopes of the regret against $\ln t$ between $t = 500$ and $t = 5000$ are 5.5 and 0.9, a similar ratio. They are below the constants, as is common at finite T, since the Lai–Robbins bound is asymptotic.

Extra experiment. Normal–Normal TS was also run on the Bernoulli rewards, i.e. with a misspecified likelihood. It still finds the best arm, but its regret (33.2) is about the same as on the noisy Gaussian bandit and four times that of Beta–Bernoulli TS. Because it assumes $\sigma^2 = 1$, its posteriors shrink as $1/N_i$ instead of as $\mu(1-\mu)/N_i$. They stay too wide, so the algorithm cannot exploit the lower noise of the Bernoulli rewards. Thompson Sampling is robust to a wrong likelihood, but it is efficient only when the model matches the rewards.